

Q. How do the lead-acid battery desulphators work? Also, how do they differ from normal battery chargers?

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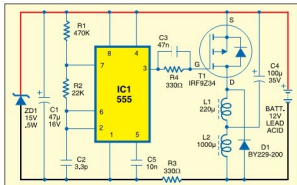
Bilasi, Deogarh, Jharkhand

A. When batteries (whether sealed lead-acid or wet batteries having liquid acid) are not charged and maintained properly, these sulphate up. Sulphation is the formation of sulphate crystals on the surface as well as inside the battery plates. It increases the electrical resistance within the battery reducing the capacity through electri-

act somewhat like an ultrasonic cleaner.

However, a desulphator will not recover batteries that have other ailments such as shorted cells, coating with other chemicals because other than distilled or de-ionised water was used to replenish the cells, and lack of acid.

The circuit of a simple desulphator is shown here in the figure. The 555 timer is connected as an astable oscillator with its output frequency set by R1, R2 and C2. Its output pulses drive the gate of MOSFET T1, which turns on to charge inductors L1 and L2. At the end of each pulse, T1 turns off and the inductors develop a high-voltage, high-current pulse which is applied across the battery via fast-recovery diode D1 and the 100µF capacitor. The 555 is protected from the high-voltage pulses via its isolated supply, by virtue of the 15V zener diode (ZD1), the 47µF capacitor and the 330-ohm resistor (R3).



A typical desulphator circuit

cal insulating properties of sulphate crystals and through the blocking of the chemical interface. These sulphate crystals disrupt the flow of electricity and act as an insulator to it. As a result, the charging and discharging capability of the battery gets destroyed. The performance of the battery thus becomes poor in due course of time. It loses its ability to supply heavy currents.

The sulphation problem can be overcome with the help of a basic battery desulphator using advanced control techniques. The battery desulphator removes the sulphate crystals. A good desulphating unit uses sharp pulses of current at about 800 kHz to set up a resonance which "jars, crushes, grinds or dissolves" sulphate crystals through internal resonance, both mechanical and electrical, wearing down the crystals so they can be recombined into the sulphuric acid of the battery electrolyte. This action appears to occur at an ionic level whereby the resonance from the desulphator may

converts soft, recently-formed lead sulphate back to its component materials (lead, lead dioxide and sulphuric acid). As batteries age, sulphate crystals become larger and harder, making conversion of lead-sulphate more difficult during recharging. Extended storage or use without a 100 per cent recharge also creates larger, harder sulphate crystals that are difficult to convert during charging. Repeated undercharging causes lead-sulphate to harden or form larger crystals on the positive plates.

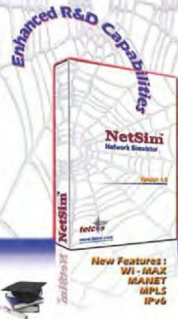
Normally, battery chargers are used for trickle charging. These restore the discharged battery. Desulphators, in comparison, are used for recovery of a desulphated battery. Single units with battery charger and desulphator combined are also available in the market.

Answers compiled by Col (Retd) N.C. Pande

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